WEEK 7 – DATA CLEANING AND TRANSFORMATION

NAME: NIVEDHA A

DEPT.: B.SC DS & AI

ROLL NO.: E25DA031

INTRODUCTION:

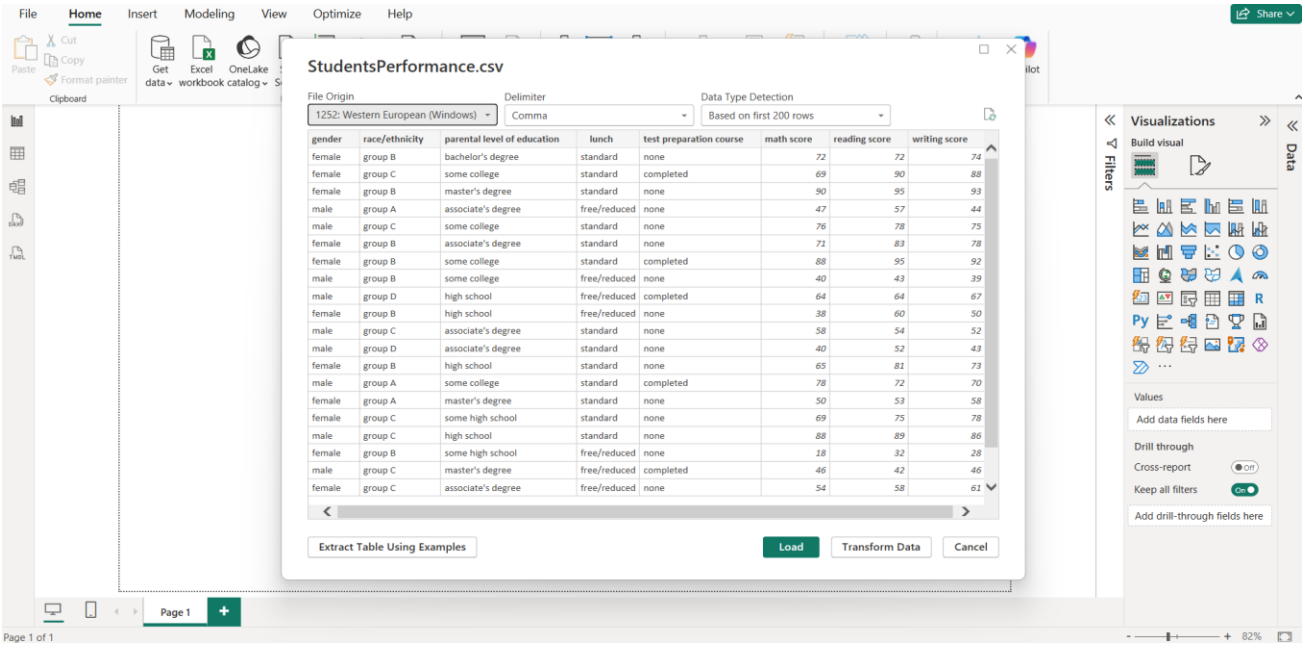
The Week 7 project focuses on cleaning, transforming, and preparing the Student Performance dataset using Microsoft Power BI. The dataset contains student background information along with Mathematics, Reading, and Writing scores. Power Query was used to clean and transform the data, while Model View and DAX were used to organize the data and create performance measures. The prepared model can be used for further student performance analysis.

OBJECTIVES:

The main objective of this project is to prepare the Student Performance dataset for effective analysis in Power BI. Student ID, Total Marks, Average, Grade, and Result were created from the available data. The subject scores were also transformed into a separate table, relationships were created between tables, and DAX measures were developed to evaluate student performance.

DATA IMPORT:

The Student Performance dataset was imported into Power BI Desktop using the Text/CSV option. After selecting the dataset, Transform Data was chosen to open Power Query Editor. The imported data was checked to make sure that the required student details and subject score columns were available before starting the transformation process.

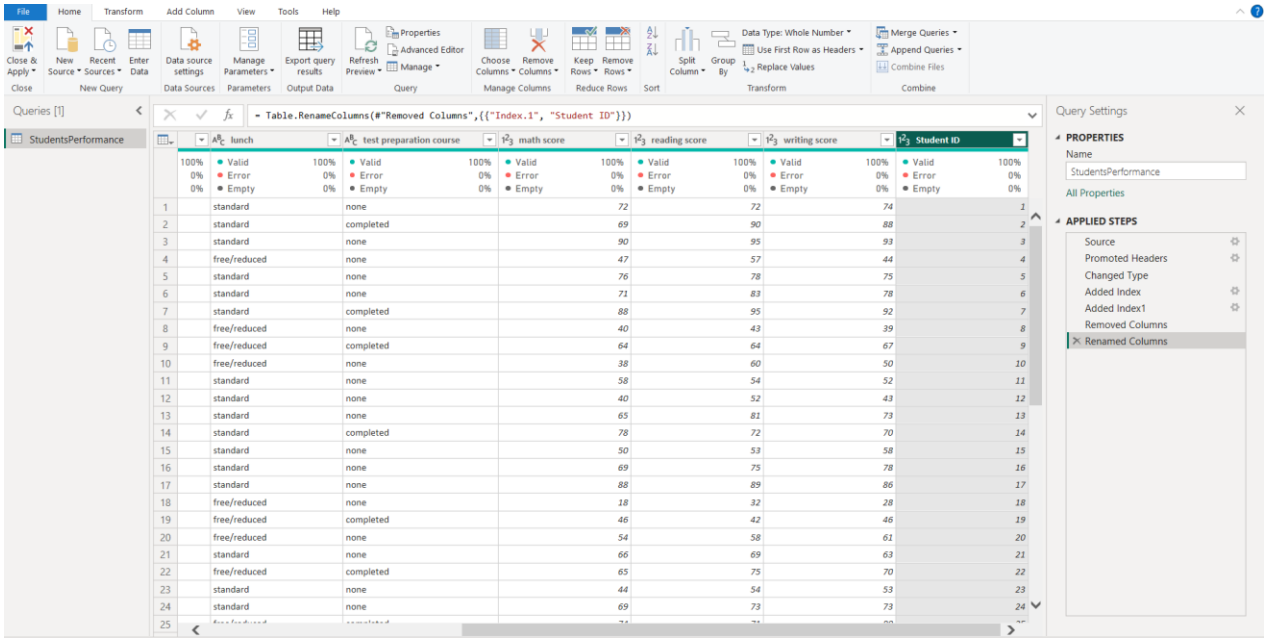


DATA CLEANING:

The text columns were cleaned and standardized using Power Query transformation options. Capitalize Each Word was used to maintain consistent text formatting, while Trim was applied to remove unnecessary spaces and Clean was used to remove unwanted characters. This helped maintain consistent values in fields such as gender, race/ethnicity, parental education, lunch, and test preparation course.

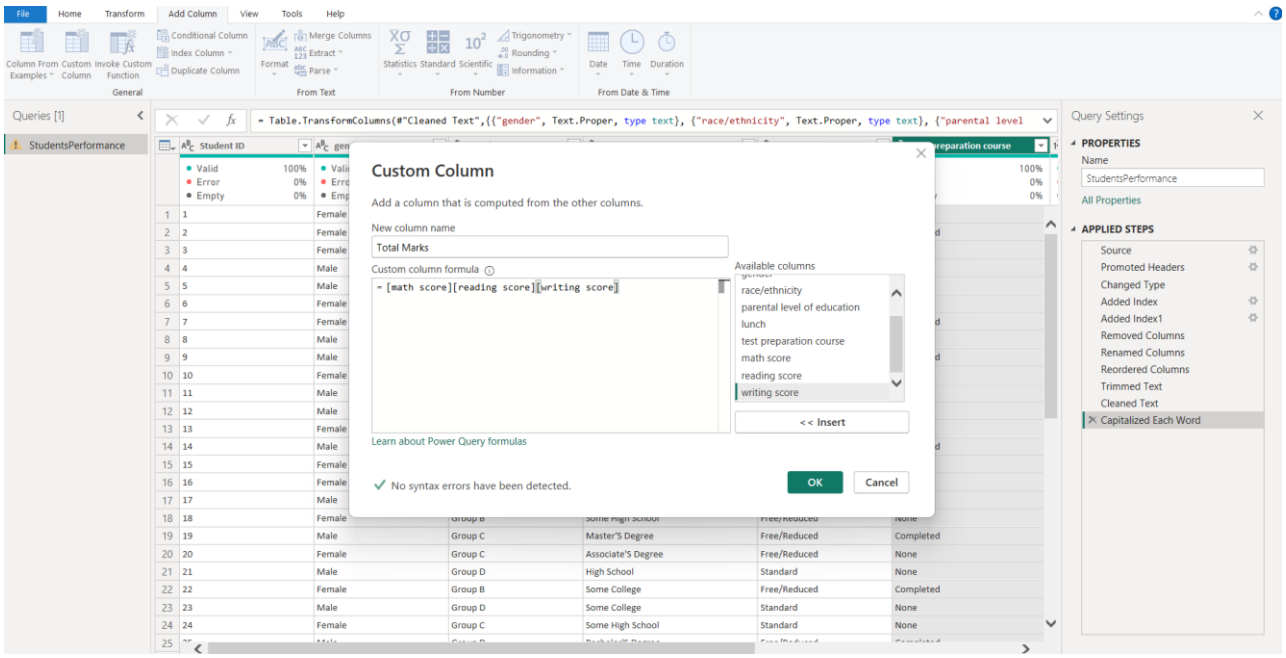
CREATING THE STUDENT ID:

A unique Student ID was created to identify each student separately within the dataset. An Index Column was added from 1 and renamed as Student id, with the data type set to Whole Number. This ID was later used as the common field for connecting the student-related tables in the data model.



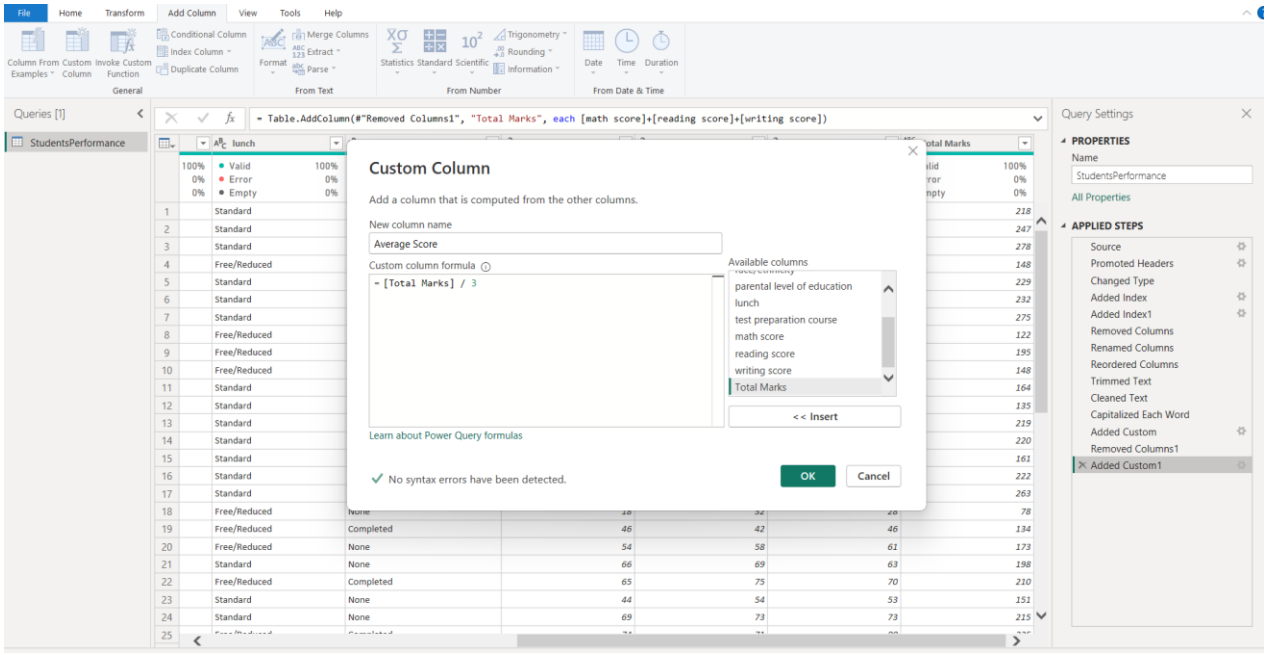
CREATING THE TOTAL MARKS COLUMN:

A Total Marks column was created using a Custom Column in Power Query. The Mathematics, Reading, and Writing scores were added together to calculate the total marks obtained by each student. The formula used was **[math score] + [reading score] + [writing score]**, and the resulting column was stored as a Whole Number.



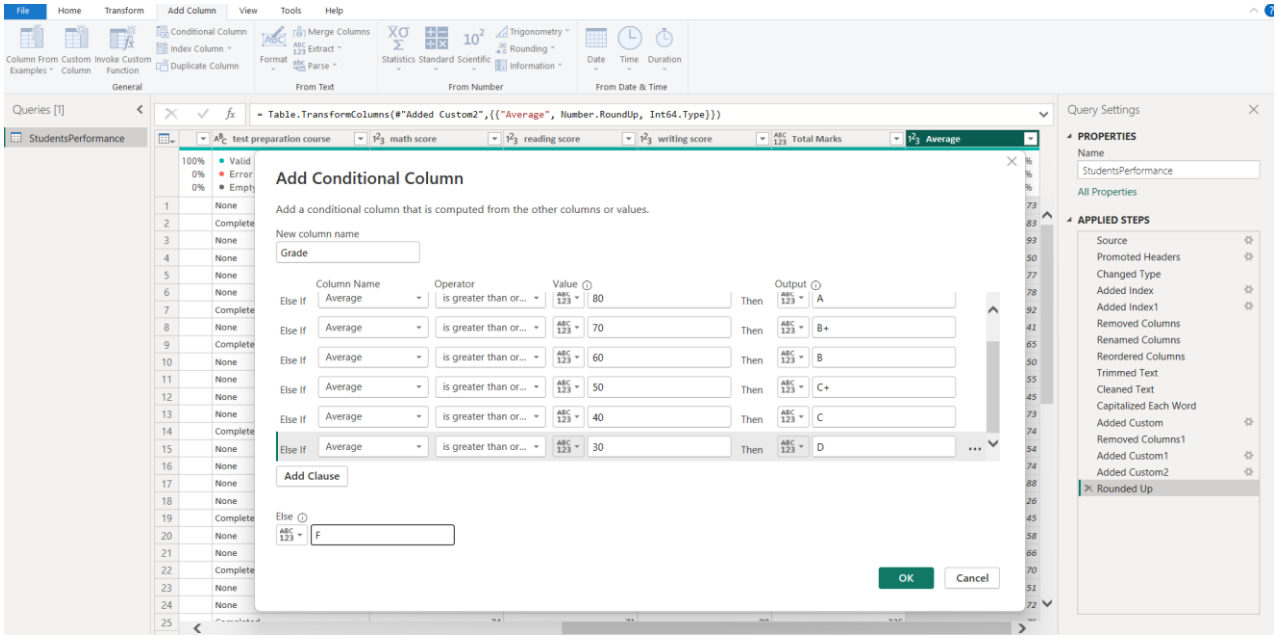
CREATING THE AVERAGE COLUMN:

An Average column was created to calculate the overall score of each student across the three subjects. The Total Marks value was divided by 3 using the formula **[Total marks] / 3**. This provides a single average score that can be used to compare the overall academic performance of students.



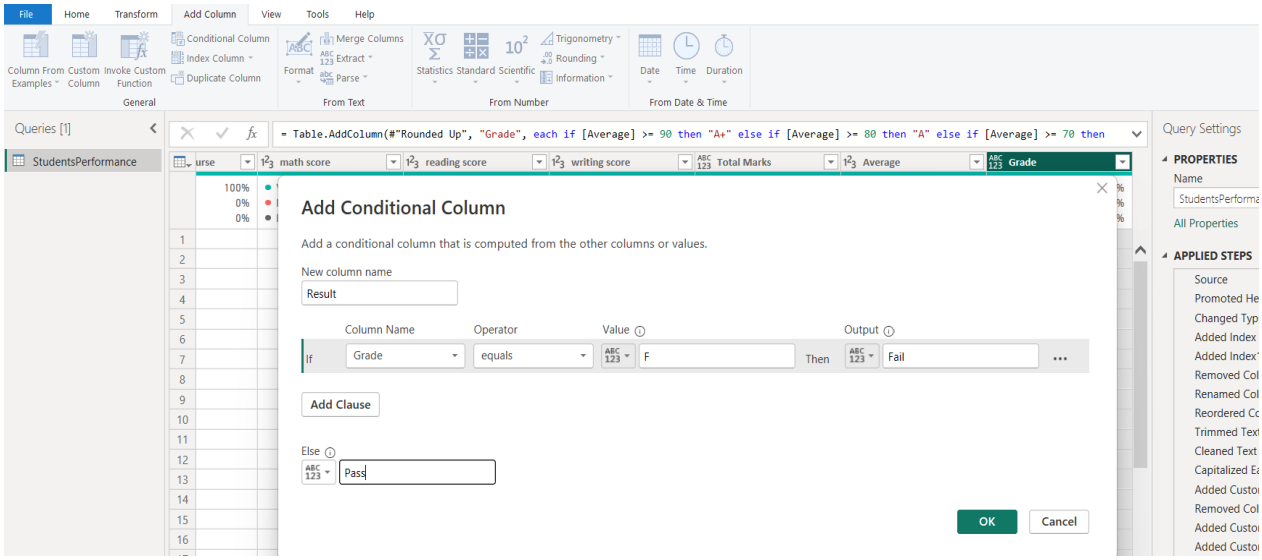
CREATING THE GRADE COLUMN:

A Grade column was created using the Conditional Column option in Power Query. Conditions were added according to the grading criteria used in the project, allowing each student to be assigned a grade based on their performance. This classification makes it easier to group students according to their academic results.



CREATING THE RESULT COLUMN:

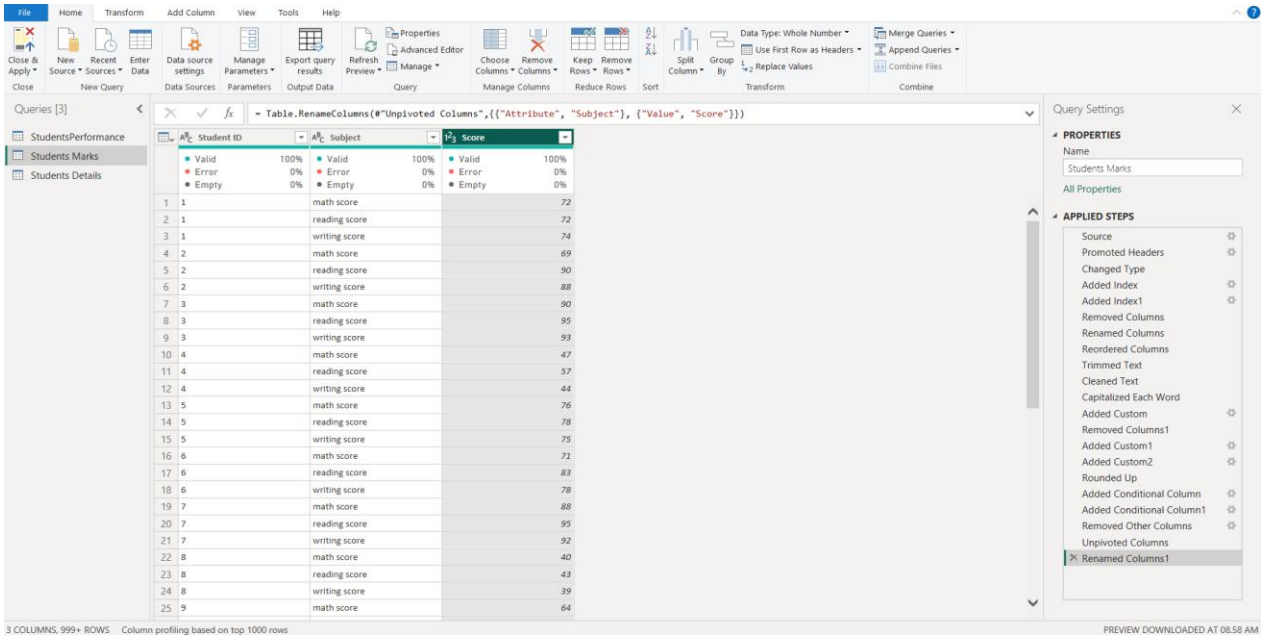
A Result column was created to classify students as Pass or Fail. The condition used in the model was that students with Grade F were classified as Fail, while students with other grades were classified as Pass. This column provides a simple overall result for each student and is later used in the pass percentage calculation.



DATA TRANSFORMATION AND TABLE CREATION:

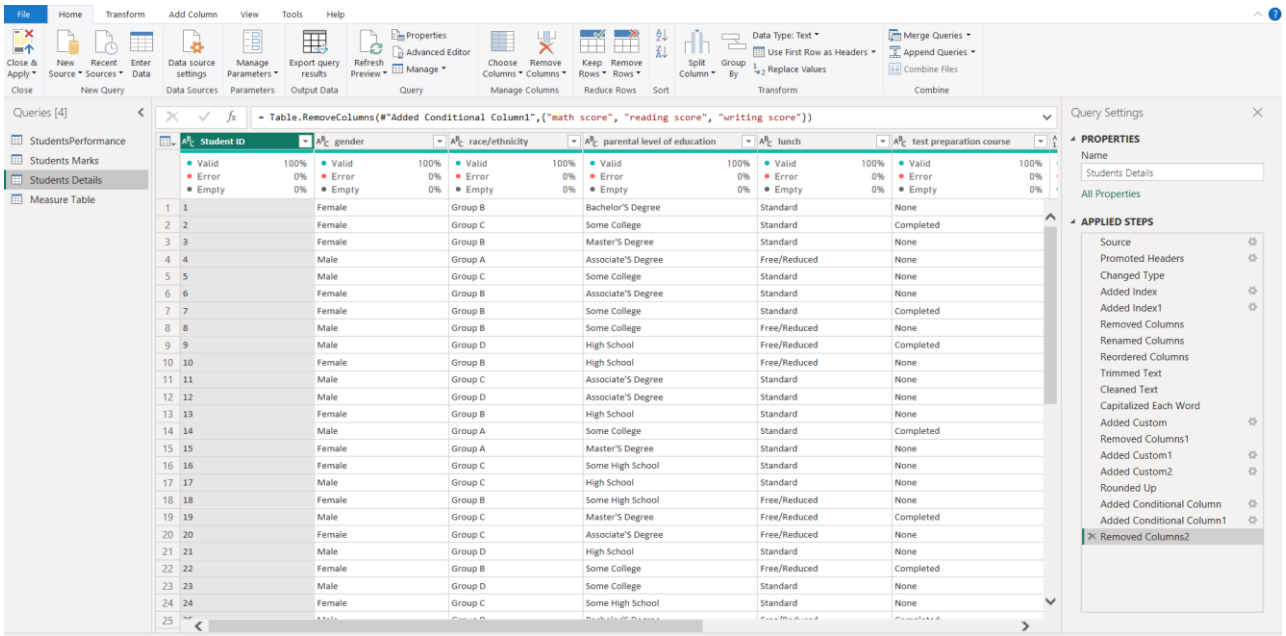
CREATING THE STUDENTS MARK TABLE:

A separate Students Mark table was created from the original dataset to organize the subject scores in a long-format structure. The main query was duplicated and only Student ID, Mathematics, Reading, and Writing scores were retained. The subject columns were then unpivoted, with Attribute renamed as Subject and Value renamed as Score.



CREATING THE STUDENTS DETAILS TABLE:

A Students Details table was created to store important student information along with their calculated performance values. The table contains Student ID, Gender, Race/Ethnicity, Parental Level of Education, Lunch, Test Preparation Course, Average, Grade, and Result. This separate table helps organize student background and performance information for analysis.



CREATING THE MEASURE TABLE:

A separate Measure Table was created to store the DAX calculations in an organized manner. The table was created using Home → Enter Data, with a column named Measure and a sample value entered. Keeping the measures in a dedicated table makes the Power BI model easier to manage and understand.

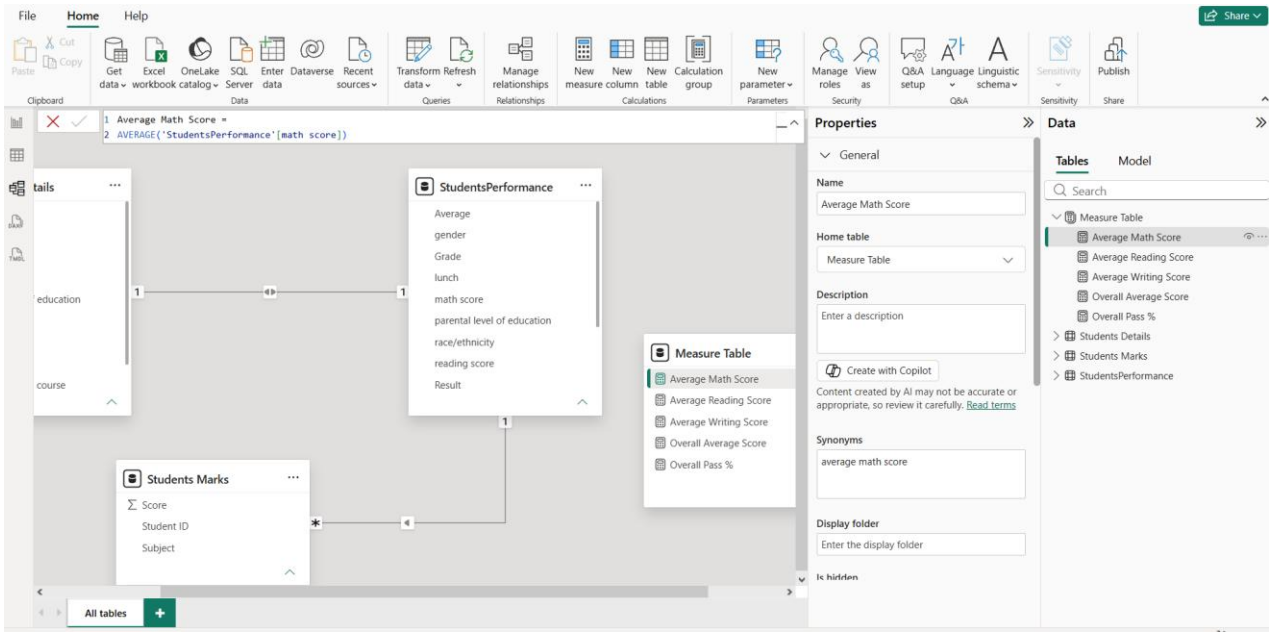
DAX MEASURES:

CREATING AVERAGE MATH SCORE:

The Average Math Score measure was created using the AVERAGE function. The formula calculates the average Mathematics score from the StudentsPerformance table. This measure provides an overall view of student performance in Mathematics.

Average Math Score =

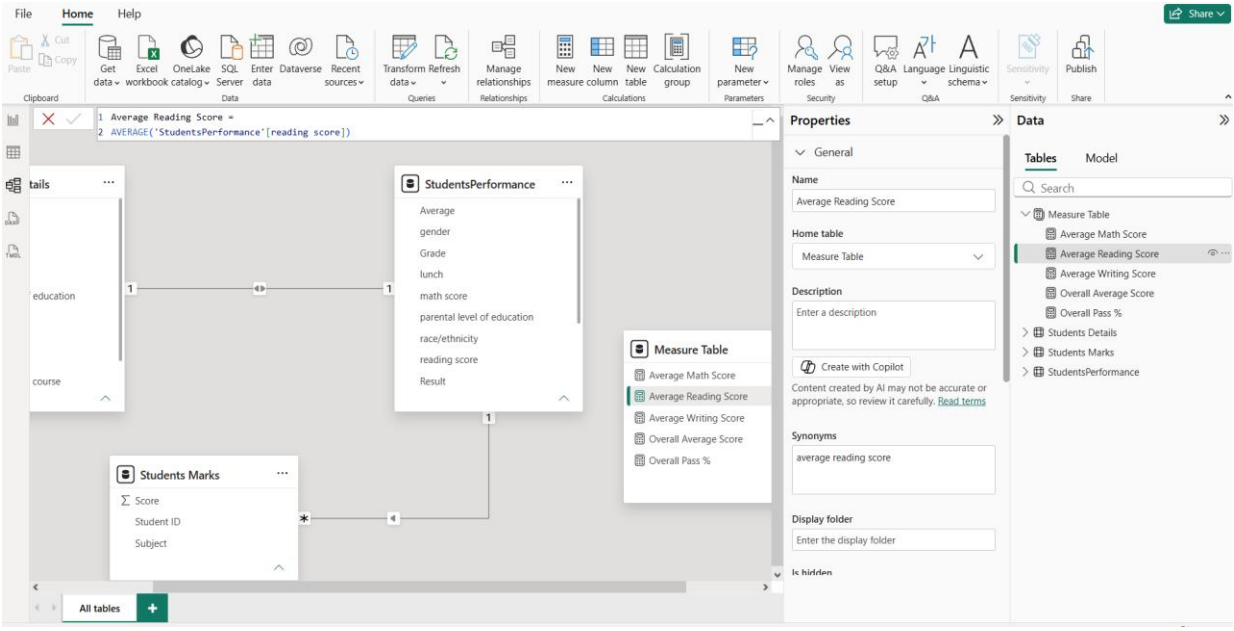
AVERAGE('StudentsPerformance'[math score])



CREATING AVERAGE READING SCORE:

The Average Reading Score measure was created to calculate the overall average score obtained by students in Reading. The AVERAGE function was applied to the reading score column from the StudentsPerformance table.

Average Reading Score = AVERAGE('StudentsPerformance'[reading score])



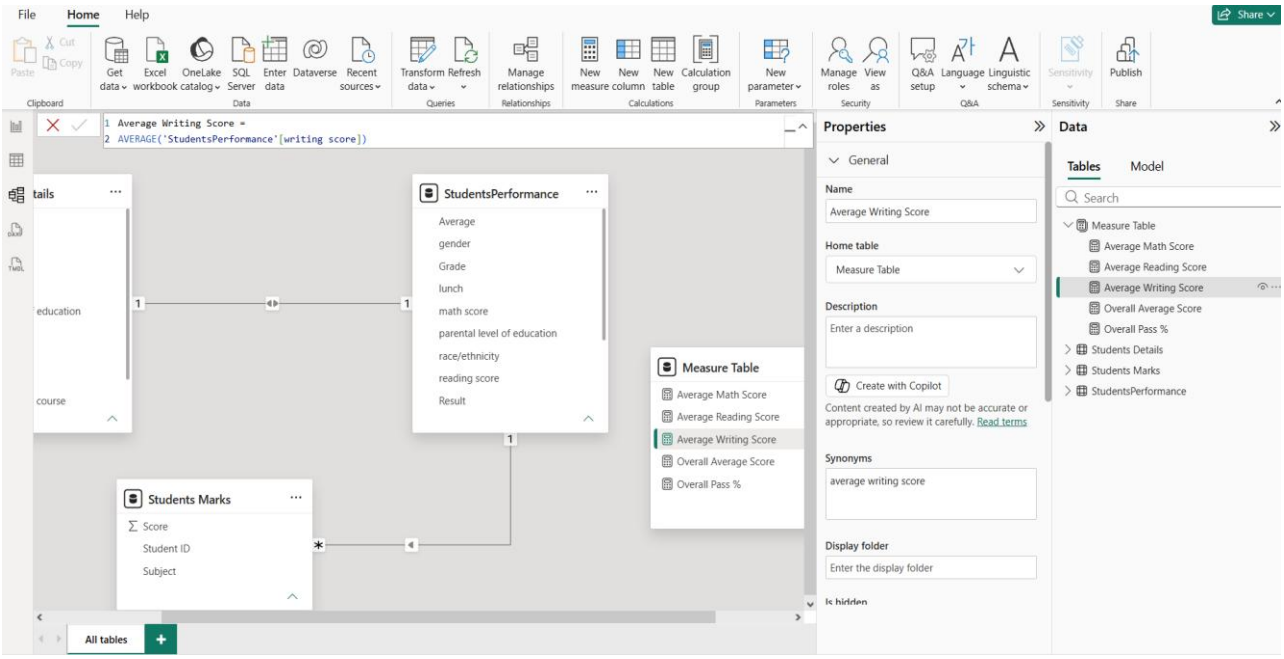
CREATING AVERAGE WRITING SCORE:

The Average Writing Score measure calculates the average Writing score of the students. This measure helps compare Writing performance with the other subjects.

Average Writing Score =

AVERAGE('StudentsPerformance'[writing score])

These three subject-wise measures are part of the five DAX measures created in the model.

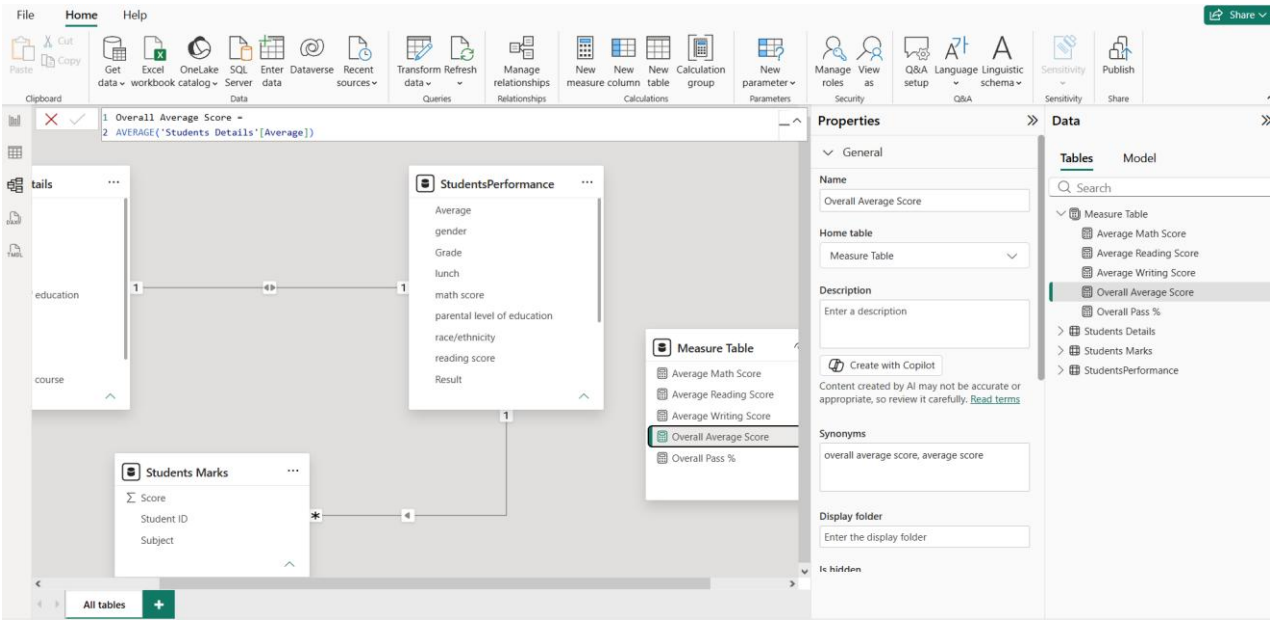


CREATING OVERALL AVERAGE SCORE:

The Overall Average Score measure was created using the Average column from the Students Details table. It calculates the overall average performance of the students and provides a single measure for evaluating general academic performance.

Overall Average Score =

AVERAGE('Students Details'[Average])



CREATING OVERALL PASS PERCENTAGE:

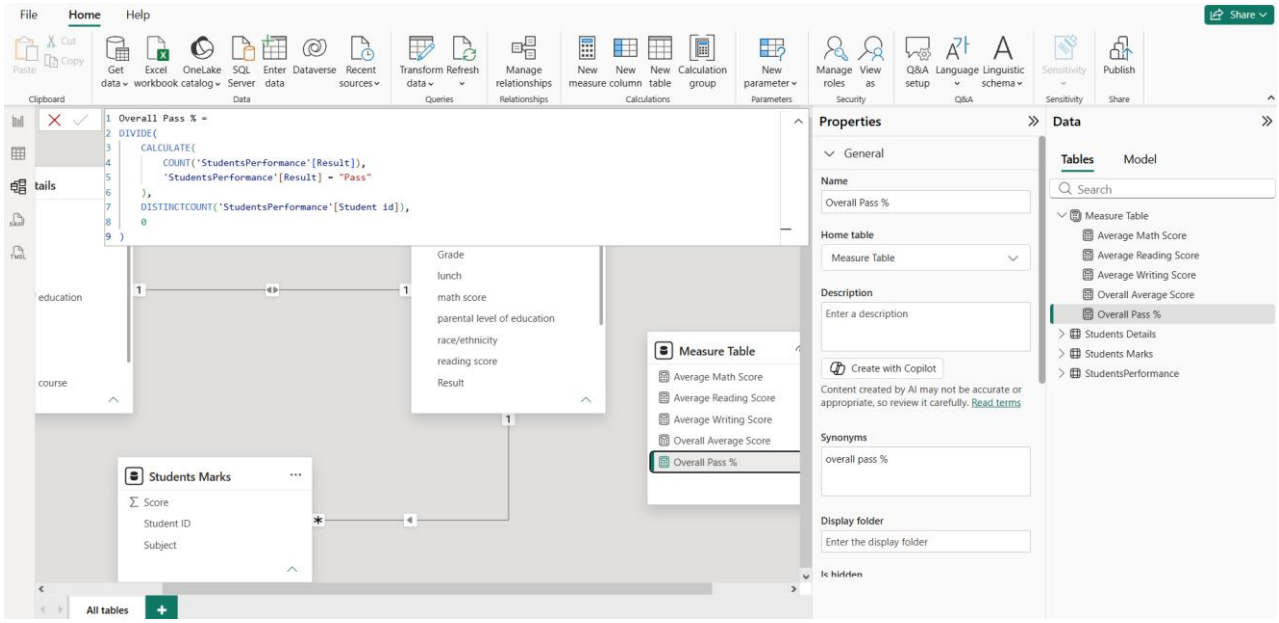
The Overall Pass % measure was created to calculate the percentage of students who passed. The CALCULATE function counts students whose Result is Pass, while DISTINCTCOUNT identifies the total number of students. DIVIDE is used to safely calculate the percentage.

```
Overall Pass % =
DIVIDE(
  CALCULATE(
    COUNT('Students Details'[Result]),
    'Students Details'[Result] = "Pass"
  ),
  DISTINCTCOUNT('Students Details'[Student id]),
```

0

)

This measure provides an overall indication of the success rate of the students.



RESULTS:

The Student Performance data was successfully cleaned, transformed, and organized in Power BI. Calculated columns were created to identify student performance through Total Marks, Average, Grade, and Result. Separate Students Mark and Students Details tables were also prepared, while DAX measures provided subject-wise averages, overall average performance, and pass percentage for further analysis.

CONCLUSION:

The Week 7 project successfully demonstrated the process of preparing student performance data using Power BI. Power Query was used for cleaning, calculated columns, and data transformation, while Model View was used to organize the relationships between tables. DAX measures were created to evaluate student performance, providing a structured data model that can be used for further visualization and analysis.